

Alessandro Puddu

Senior Unity Engineer | Games, VR & Real-Time 3D

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PROFILE

5+ years of professional Unity/C# experience, including contributions to the commercial PC game *Alaloth: Champions of the Four Kingdoms*, VR development on *RaceClub* for **Sandbox VR**, enterprise R&D at Capgemini Engineering, and contract technical leadership at Squircle Games. Hands-on across gameplay, performance, WebGL/VR porting and **LLM and speech integration**; extensive use of **coding agents** within reviewed and tested development workflows.

PROFESSIONAL EXPERIENCE

Solid Code Interactive

Independent Senior Unity Engineer

December 2025 - Present

Remote / International

Squircle Games - Unity Lead (Contract)

March 2026 - Present

Technical direction and hands-on delivery across Unity architecture, gameplay, networking, XR, tooling and connected services.

- **Integrated LLM, TTS and STT** with character interactions, streamed text/audio and conversational state. Built browser/native speech implementations and client-backend authentication and session flows across desktop, WebGL and VR, including cancellation and error recovery.
- **Implemented Meta Quest VR/MR interactions**: passthrough, hand tracking, controller/hand switching, locomotion and spatial UI using OpenXR and XR Interaction Toolkit. Optimised standalone rendering through platform-specific graphics settings, shaders and resource management.
- **Built character systems and authoring tools**: animation, inverse kinematics, physics-driven reactions, camera behaviour and editor tooling for content creation.
- **Established coding-agent workflows** for implementation, debugging and refactoring, with repository guidance, scoped tasks, code review and automated validation of generated changes.

Sandbox VR - Unity Developer (Contract)

May 2026 - Present

RaceClub - Unity VR product

- **Integrated UI Toolkit into runtime menus and HUD systems**, connecting interfaces to gameplay state and data-driven behaviour.
- Implemented cross-device input and feedback flows, delivered gameplay features and resolved VR state issues.

Capgemini Engineering

Unity Software Engineer (R&D)

October 2023 - February 2026

Bologna / Remote

- **Refactored CPU-heavy physics and simulation systems** for real-time performance on constrained hardware, diagnosing bottlenecks and changing the affected runtime logic.
- **Decoupled asset delivery from application releases** through Addressables-based content loading, including secure on-demand access.
- Delivered confidential WebGL and VR applications with multiplayer, in-app authoring and platform-specific interactions; **ported content and runtime systems across WebGL, mobile and VR**.
- Built node-based editor tools and adapted persistence to browser constraints.

PROFESSIONAL EXPERIENCE (CONTINUED)

Gamma Interactive

March 2021 - September 2023

Game Programmer

Remote

Alaloth: Champions of the Four Kingdoms - Commercial PC Action-RPG [Steam]

- **Delivered production gameplay and UI features** for *Alaloth*, including inventory and skill systems, gameplay loops, narrative events and UI architecture.
- **Replaced repetitive manual production operations** with in-engine tools for asset validation and world management.
- Resolved production issues in collaboration with design and QA while supporting ongoing feature development.

TECHNICAL SKILLS

- **Unity & platforms:** Unity, C#, PC, WebGL, mobile, Meta Quest, OpenXR, XR Interaction Toolkit, UI Toolkit, Addressables, cross-platform porting, profiling and optimisation.
- **Gameplay & architecture:** NPC/game AI, state machines, physics, simulation, animation, IK, shaders, async workflows, persistence, editor tooling, diagnostics and testing.
- **AI integration:** LLM-backed interactions, conversational state/context, TTS/STT, microphone capture, streamed audio, browser/native speech interfaces.
- **Networking & full stack:** Unity Netcode, Mirror, REST, WebSockets, SSE, authentication, TypeScript, Node/Express, Next.js, Prisma, PostgreSQL.
- **AI-assisted engineering:** Coding agents, context engineering, task decomposition, repository instructions, code review, automated validation and technical documentation.

ADDITIONAL PROJECTS & RECOGNITION

Apocalypse Rangers

2020 - 2021

Gameplay & Network Programmer - PC / Online [Trailer]

- Built Mirror-based client-server multiplayer and custom HLSL toon/terrain rendering. Project awarded **3rd Place Online, New Game Designer 2020**.

Hell for One

2019 - 2020

AI & Gameplay Programmer - PC / RTS Hybrid [Itch.io]

- Implemented NPC decision logic, state machines, steering, reusable actions and battle spawning/coordination. Project awarded **1st Place Offline, New Game Designer 2020**.

RuNeglect

2018 - 2019

Sole Developer - Microsoft HoloLens / AR thesis

- Built a rehabilitation application using spatial mapping, object pooling and room-adaptive gameplay.

EDUCATION

B.S. in Computer Science, Università degli Studi di Milano

2019

Thesis: AR rehabilitation for hemispatial neglect using Microsoft HoloLens.

Master's coursework in Computer Science

2019 - 2021

Università degli Studi di Milano. Advanced graphics, real-time rendering, AI and game development.

LANGUAGES

Italian: Native | **English:** Professional working proficiency